Related work: active inference, federated Bayes, and the scoped bridge I

This work sits at the boundary between two research communities with different objects of inference. Each cited thread contributes a real component; this paper extends the intersection without claiming that either literature is exhausted. The positioning is therefore against the reviewed sources and their assumptions, not against an absolute claim about everything the field has or has not done.

The pre-modern sources in this manuscript are not evidence that early probability theorists anticipated KL minimization or federated learning. They serve a narrower role: they show that the paper's recurring problems — expected uncertain outcomes, inverse inference from effects to causes, utility-sensitive action, and collective judgment — are old problems now implemented with modern variational machinery.

The probability-calculus line begins with Pascal and Fermat's correspondence on the problem of points (Pascal and Fermat 1654), Huygens' printed treatment of reasoning in games of chance (Huygens 1657), Montmort's combinatorial analysis of games (Montmort 1708), Bernoulli's *Ars conjectandi* and its law-of-large-numbers logic (Bernoulli 1713), and de Moivre's systematic probability textbook (Moivre 1718). For this paper, their relevance is not that they contain active inference, but that they make expectation and uncertain evidence objects of calculation.

The inverse-probability line is closer to the manuscript's formal spine. Bayes and Price frame the problem of inferring an unknown chance from observed events (Bayes 1763), and Laplace generalizes the probability of causes given events (Laplace 1774). Read beside modern active inference, these sources support the vocabulary of generative-model inversion: data are effects, latent states or parameters are causes, and a posterior belief reconciles prior commitments with observed evidence. That is a conceptual bridge only; the actual identity with the FedGVI recovery corner is the modern result of sec. 5.8 and sec. 6.

Decision and aggregation enter through Daniel Bernoulli's expected-utility treatment of risk (Bernoulli 1738) and the voting-theoretic work of Borda and Condorcet (Borda 1784; Condorcet 1785). They help name the problem this paper revisits in a categorical active-inference colony: local judgments must be combined, and the chosen aggregation rule encodes assumptions about competence, independence, weights, and stakes. Those assumptions are exactly what the modern log-pool and robustness literature make explicit.

Active inference: generative agents, EFE, and colonies I

The free-energy principle (Friston 2010) and its discrete state-space process theory (Friston et al. 2017) gave the field a unified account of perception and action: hand-built generative models with $A/B/C/D$ matrices, variational free energy for inference, and expected free energy for principled action selection. The discrete state-space synthesis (Da Costa et al. 2020) and toolboxes turned that account into scalable machinery — the pymdp discrete-state library (Heins et al. 2022) and the RxInfer reactive message-passing engine (Bagaev et al. 2023). *Yes:* this substrate is exactly what we reimplement, and our EFE decomposition into risk and ambiguity (eq. 18, eq. 19, fig. 7) is the community's own action-selection objective. *And:* we add a robustness knob to its belief-fusion step without leaving the framework.

The fusion operator itself is not new as a mathematical object. Under the explicit categorical posterior-log-potential and fixed-weight bridge of sec. 5.8, the message-combination term of Friston's belief-sharing equation is represented by a logarithmic opinion pool (Genest and Zidek 1986; Genest et al. 1986) and a product-of-experts consensus (Hinton 2002). That limited representation is not a statement that the complete source protocol is a log pool. Abbas' KL view of linear and log-linear pools makes the same point in scoring-rule language: log-pooling can be justified as a KL aggregation rule for expert distributions, not as a contamination-robust estimator (Abbas 2009). The Bayesian Committee Machine adds the distributed-learning analogue: independent estimators trained on data subsets can be combined by a product-style Bayesian rule, with assumptions about conditional independence and prior accounting made explicit (Tresp 2000). Fully Bayesian aggregation gives the social-choice counterpart: geometric pooling of beliefs is singled out by dynamic Bayesian rationality conditions, not by robustness to contaminated reports (Dietrich 2021). That classical literature is useful precisely because it names the hidden commitments — shared support, weight choice, prior accounting, and external-Bayes coherence — that are easy to overlook when the same operation appears as a colony update.

A parallel thread studies how active-inference agents coordinate as ensembles — collective behavior and surprise minimization across a colony (Heins et al. 2024), epistemic communities (Albarracin et al. 2022), and collective intelligence (Kaufmann et al. 2021). The belief-sharing thread (Friston et al. 2024) is one member of this family: agents reach consensus by exact-Bayes fusion of one another's beliefs, and the colony belief-sharing scenario we implement as a reduced categorical standard-Bayes-limit analogue (eq. 10, sec. 7.2) is related to its worked example. Structure learning rounds out the picture: active inference with Bayesian optimal design selects and reduces models within the same frame (Smith et al. 2022), and post-hoc Bayesian model reduction (Friston and Penny 2011) is the engine behind our emergence study (eq. 16, sec. 7.4). The cited active-inference line has rich generative models, principled action, and ensembles that coordinate by sharing beliefs. In the sources reviewed here, the belief-fusion rule is not systematically characterized under explicit contamination or intentionally wrong broadcasts, nor connected to the robustness theory used here. Fusion is treated as exact-Bayes and trusting in that scoped comparison.

Friston et al. (2024) crystallized these ideas into three worked simulations: (1) communicating-colony free-energy convergence, (2) Dirichlet language acquisition, and (3) Bayesian model reduction structure emergence. We use reduced categorical standard-Bayes-limit analogues of all three mechanisms (sec. 7.1) under the declared protocol before building the robust extension. This is a mechanism-level comparison, not an exact source-protocol or figure reproduction.

Federated learning aggregates models trained on decentralized data (McMahan et al. 2017), and the probabilistic-federation line recasts that aggregation as variational inference — partitioned variational inference (Ashman et al. 2022) and its federated predecessor (Bui et al. 2018). This is a different use of the word *federated* from Friston et al.'s federated inference: Friston federates hidden-state beliefs among agents sharing a world model, while machine-learning federated learning usually federates parameter or predictive-model updates over decentralized datasets. The bridge claimed here is therefore algebraic and variational — a shared normalized product/log-pool operator at the KL/NLL recovery corner — not a claim that either source paper already solved the other's problem.

Robustness, meanwhile, has a mature Bayesian theory: general Bayesian updating through a loss (Bissiri et al. 2016), Gibbs posterior inference (Jiang and Tanner 2008), safe learning-rate selection under misspecification (Grünwald 2012), coarsened posteriors for robustness to exact-data conditioning (Miller and Dunson 2018), divergence-criteria posterior updating (Jewson et al. 2018), Bayesian misspecification asymptotics (Kleijn and Vaart 2012), the optimization-centric generalized variational inference view (Knoblauch et al. 2022), recent closed-form characterizations of unrestricted generalized variational objectives (Nguyen and Westerhout 2026), and the bounded-influence losses that make updates robust — density-power and gamma-divergence estimation (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015), robust-divergence variational inference (Futami et al. 2018), and generalized cross-entropy (Zhang and Sabuncu 2018). Huber and Ronchetti supply the robust-statistics vocabulary of influence and breakdown (Huber and Ronchetti 2009), which we use as vocabulary for boundedness and empirical failure modes rather than as a theorem transfer. FedGVI (Mildner, Hamelijnck, et al. 2025) is the synthesis we use per agent: a robust generalized- Bayes objective with client and server divergence choices. This community provides decentralized aggregation and theorem-backed results in its stated settings. The cited papers do not evaluate active-inference POMDP belief consensus — the generative-model-bearing, action-selecting setting where beliefs drive behavior.

Robust and federated Bayes outside active inference I

The 2025 preprint on convergence rates under prior misspecification (Mildner, Giampouras, et al. 2025) sharpens the current GVI context: bounded divergences can support concentration and rates under explicit assumptions, but the result does not transfer automatically to this repository's finite categorical state space or to its server-side aggregation heuristic.

The federated-learning robustness literature also supplies important negative space for this paper. Byzantine-tolerant gradient aggregation (Blanchard et al. 2017), geometric-median robust aggregation (Pillutla et al. 2022), and divergence-weighted gamma-mean aggregation (Li et al. 2022) attack corrupted client updates directly. A recent Bayesian robust-aggregation preprint likewise models unknown client honesty for federated model updates (Karakulev et al. 2025). Those are close comparators for adversarial federation, but the object being aggregated differs: they aggregate model-update vectors or posterior measures, while this paper attacks categorical belief fusion and generalized-Bayes client updates. Robust subset-posterior combination is another nearby Bayesian route (Minsker et al. 2017), but it combines posterior measures across data shards rather than active-inference belief broadcasts with a shared latent state. We use those sources to position the problem, not to import their guarantees into `robust_aggregate`.

The specific bridge added here I

The gap this manuscript fills is robust, generalized-Bayes belief fusion for active-inference ensembles. Concretely:

The specific bridge added here I

- **Per-agent bounded-influence updates.** FedGVI-faithful β/rcce updates that carry provable bounded influence (eq. 4, eq. 5).
- **Server-side divergence-reweighting heuristic.** A complementary aggregation heuristic (eq. 7) with a stated recovery limit, positioned alongside robust federated aggregation work without borrowing its theorems.
- **Provable recovery of the standard-Bayes client limit.** The client KL/NLL loss limits recover Bayes, and the separate zero-robustness server identity returns the project log-linear pool (eq. 6, eq. 3, eq. 7). Under the stated categorical bridge, that pool specializes only the Eq. 7 message-combination term (Friston et al. 2024).
- **Additional executed studies beyond the Friston et al. (2024) baseline.** The three Friston simulations motivate source-mechanism analogues rather than serving as source-protocol recovery checks; beyond them we contribute: (4) a contamination sweep in which a fraction of colony members are adversarial or misspecified (sec. 7.5); (5) a moving-world scenario with active expected-free-energy movement (sec. 13.7); (6) a two-level hierarchical POMDP (sec. 13.9); (7) a three-level hierarchical POMDP (sec. 13.11); (8) a sensitivity sweep over acuity $\times$ colony size (sec. 13.13); and (9) finite-grid parameter recovery for the declared acuity family (sec. 14).

The specific bridge added here II

- **Rigorous statistics.** Every verdict is produced by matched-pairs Wilcoxon signed-rank tests (Wilcoxon 1945; Fay and Proschan 2010) deflated with Benjamini–Hochberg FDR (Benjamini and Hochberg 1995), reported with bootstrap confidence intervals (Efron and Tibshirani 1993), rank/effect-size caveats (Nakagawa and Cuthill 2007), and an observed-effect design-power approximation — none of which appear in Friston et al. (2024).
- **An objective-backed server-side aggregator with redescending weights.** The `variational_aggregate` rule descends a stated aggregation free energy (eq. 8) monotonically; any converged fixed point is coordinatewise stationary. The rule carries a proven raw effective-weight bound and empirical redescending response (fig. 15), while leaving open whether the sharper reverse-KL server heuristic is itself the closed-form minimizer of an equally defensible objective.

The specific bridge added here I

The contribution is the bridge: an explicit connection between active-inference belief consensus and robust federated generalized Bayes, with the standard pool recovered exactly at the corner and the additional studies (the contamination sweep plus several structural extensions) showing how the bridge behaves beyond the Friston et al. baseline.